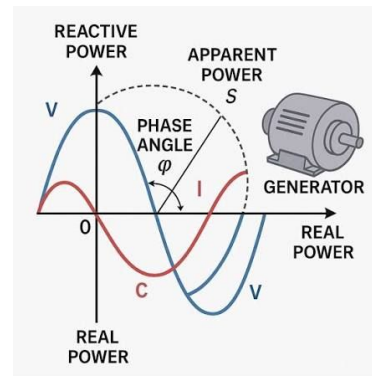
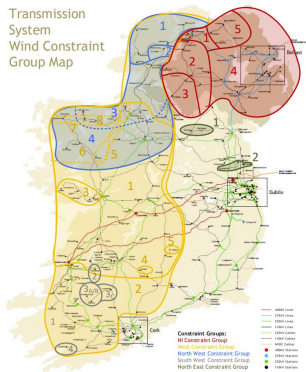


How the Grid Works

Key Topics: The DC Power Flow Approximation, Transmission System Line Ratings, A System-Level view of the problem, Interesting Problem Areas





Purpose of this talk:

- Outline the core physics behind the problem of constraint
 - Explain relevant context for the Irish grid
 - Highlight some interesting problem areas
- 

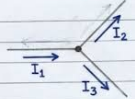
The DC Power Flow Approximation

KIRCHHOFF'S LAWS

① KCL (Kirchhoff's Current Law)

At a **JUNCTION**, total current **entering** = total current **LEAVING**.

$I_1 = I_2 + I_3$



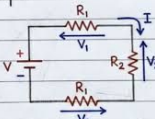
No current is lost at junction.

② KVL (Kirchhoff's Voltage Law)

In a **CLOSED LOOP**, sum of **VOLTAGE** rises = sum of voltage drops.

$V = V_1 + V_2$

(or) $\sum V = 0$

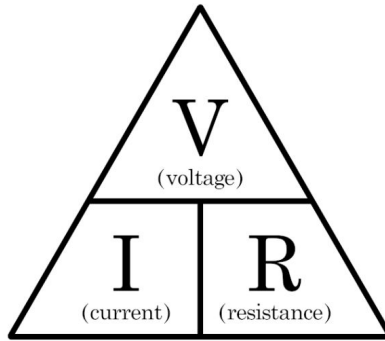


Total voltage is shared in the circuit.

Example: If 6V battery is used, voltage drop across resistors adds to 6V.

KCL: Current at junction
KVL: Voltage in loop

Ohm's Law Triangle



V
(voltage)

I
(current)

R
(resistance)

$V = I \times R$

$I = V \div R$

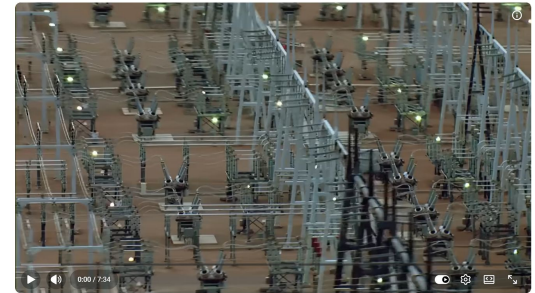
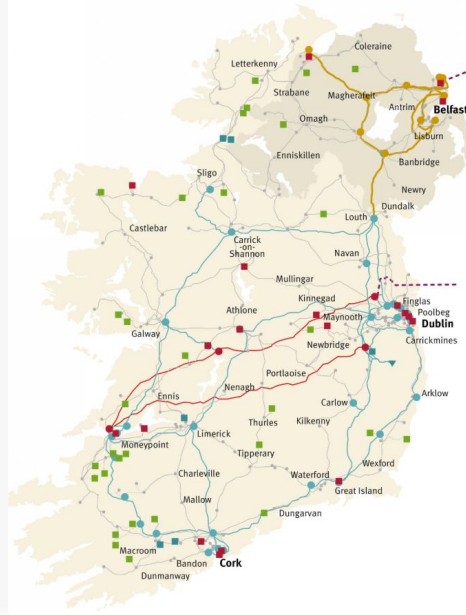
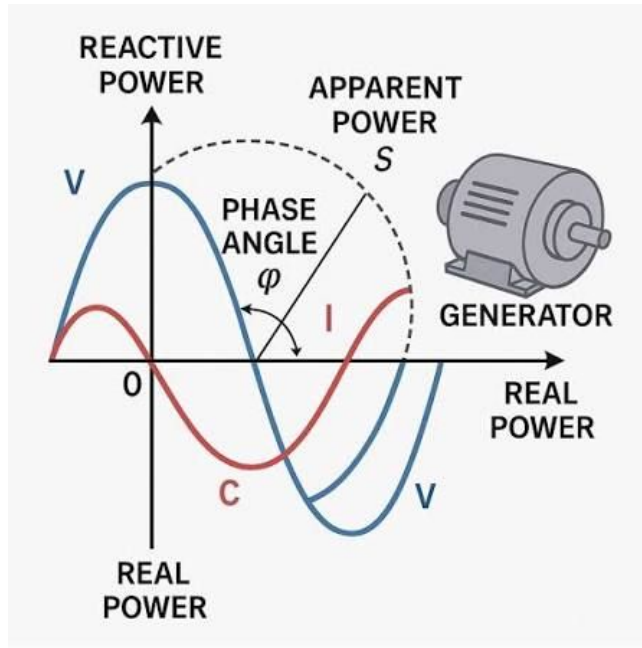
$R = V \div I$

www.inchcalculator.com

You are probably familiar with some physics for circuits and power flow

The DC Power Flow Approximation

Unfortunately, these do not apply to our alternating current (AC) grid



Thomas Edison Did Everything He Could To Stop Nikola Tesla Succeeding | Tesla's Death Ray
Discovery UK
4.5M subscribers

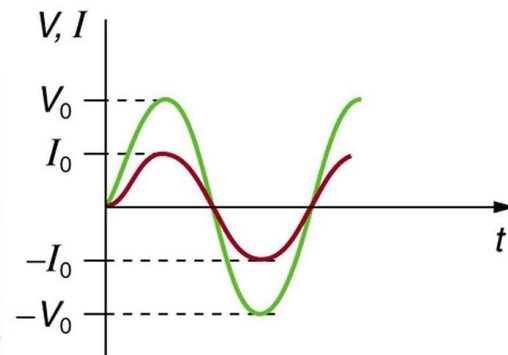
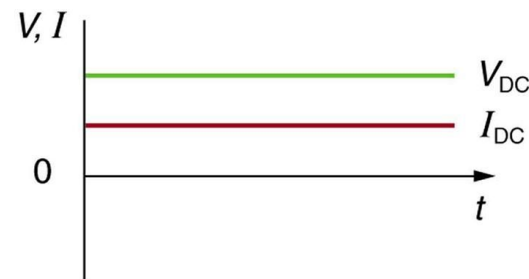
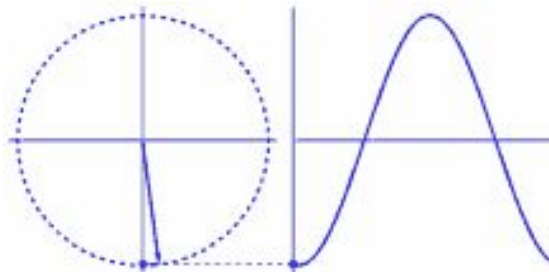
See [video](#) on interesting story behind why AC was chosen for transmission

The DC Power Flow Approximation

So... our power is in AC

We now have:

- Real Power (P) measured in Watts
-This is the power that actually does the work
- Reactive Power (W) in VAR (volt-amps reactive)
-Messy power that swashes around and does no work
- Apparent Power (S) in VA (Volt Amps)
-This is Real power + Reactive Power



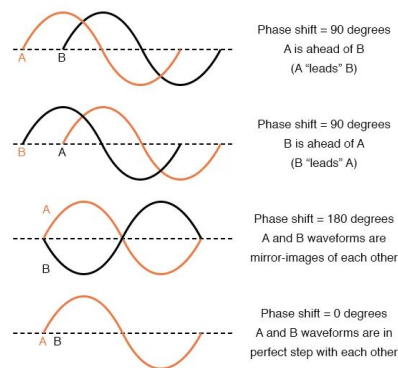
The DC Power Flow Approximation

AC power gets complicated because of this...

For a transmission line between buses i and j with series impedance $Z_{ij} = R_{ij} + jX_{ij}$, the real power flowing from $i \rightarrow j$ is:

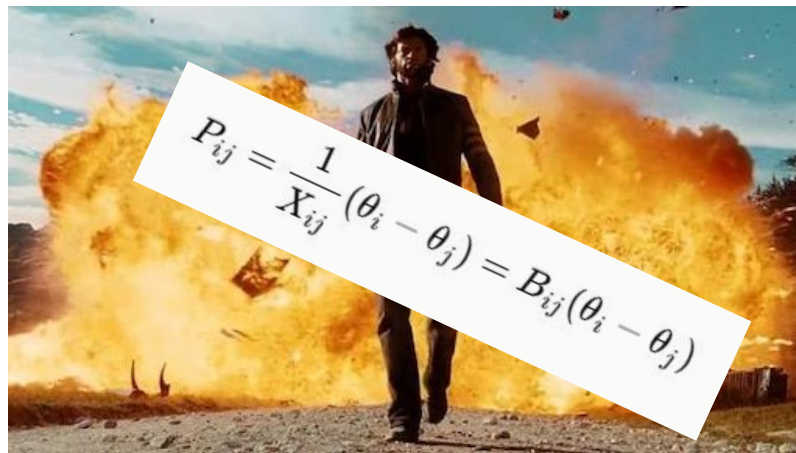
$$P_{ij} = \frac{|V_i||V_j|}{X_{ij}} \sin(\theta_i - \theta_j) + (\text{terms involving } R_{ij}, |V_i|^2, |V_j|^2)$$

Symbol	Meaning
P_{ij}	Real power flowing on the transmission line from bus i to bus j (MW)
V_i, V_j	Complex voltage phasors at bus i and bus j
$ V_i , V_j $	Voltage magnitudes at bus i and bus j (in per-unit or kV)
X_{ij}	Reactance of the line connecting bus i and bus j (Ω , or per-unit)
θ_i, θ_j	Voltage phase angles at bus i and bus j (radians or degrees)
R_{ij}	Resistance of the line connecting bus i and bus j , the term DC power flow drops by assuming $R_{ij} \ll X_{ij}$



The DC Power Flow Approximation

So let's just ignore it - The DC Power Flow Approximation baby!!



^ Me walking away from AC power flow equations because I have the DC power flow approximation and made 3 key assumptions

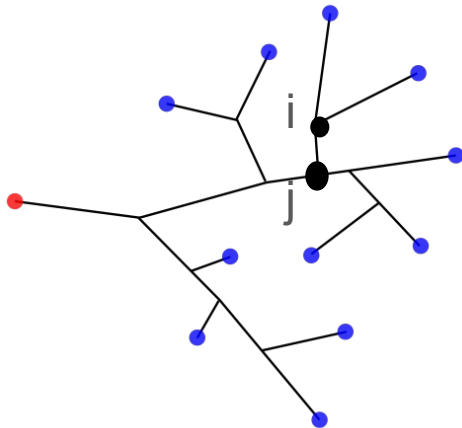
The DC Power Flow Approximation

Make three key assumptions, for transmission line between buses i and j

1. Assume $V_i = V_j$ p.u. (per unit, as a ratio of its nominal voltage)

Eirgrid works hard to keep voltages across the transmission network roughly the same, so let's assume it's perfect

If i is 110kV bus and j is 220kV, we assume both are at exactly 110kV and 220kV each (or 1 p.u.)



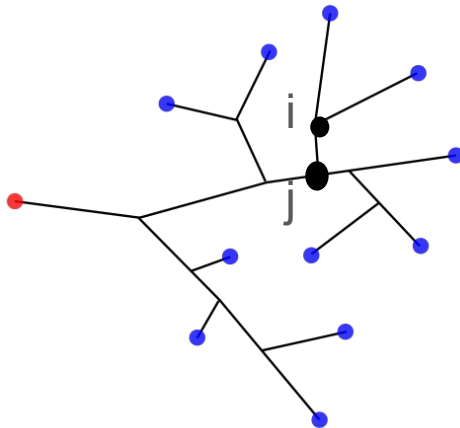
The DC Power Flow Approximation

Make three key assumptions, for transmission line between buses i and j

2. High X/R ratio $R_{ij} \ll X_{ij}$

X = Reactance, R = Resistance

Transmission lines (especially at higher voltage) are much more reactive than resistive. Basically, assume all losses come from reactance, not resistance across a line. This is a shaky assumption to make at lower voltages, especially 110kV and below

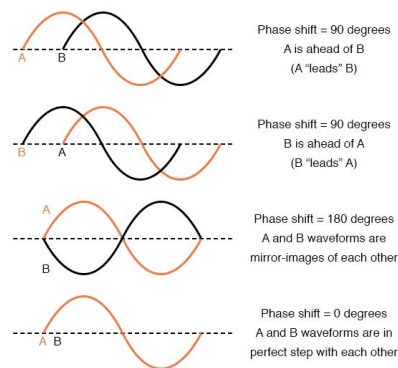
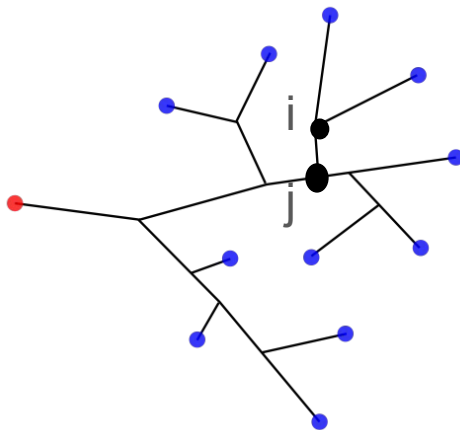


The DC Power Flow Approximation

Make three key assumptions, for transmission line between buses i and j

3. Assume small voltage phase angle differences, i.e. θ_i is roughly the same as θ_j

Under normal operating conditions, voltage angle differences across a line are small. This allows us to approximate $\sin(\theta_i) = \theta_i$



The DC Power Flow Approximation

With those assumptions, this:

$$P_{ij} = \frac{|V_i||V_j|}{X_{ij}} \sin(\theta_i - \theta_j) + (\text{terms involving } R_{ij}, |V_i|^2, |V_j|^2)$$

Simplifies to this:

$$P_{ij} = \frac{1}{X_{ij}} (\theta_i - \theta_j) = B_{ij} (\theta_i - \theta_j)$$

The DC Power Flow Approximation

DC Power flow approx.

$$P_{ij} = \frac{1}{X_{ij}} (\theta_i - \theta_j) \iff$$

DC Circuit Equivalent

$$I_{ij} = \frac{1}{R_{ij}} (V_i - V_j)$$

Symbol	Meaning
P_{ij}	Real power flowing on the line from bus i to bus j (MW)
X_{ij}	Reactance of the line connecting bus i and bus j
θ_i, θ_j	Voltage phase angles at bus i and bus j

DC circuit	DC power flow
Current I_{ij}	Real power P_{ij}
Voltage V_i, V_j	Voltage angle θ_i, θ_j
Resistance R_{ij}	Reactance X_{ij}
Conductance $1/R_{ij}$	Susceptance $B_{ij} = 1/X_{ij}$
Ohm's law	$P = B\theta$

The DC Power Flow Approximation

Power 

$$P_{ij} = \frac{1}{X_{ij}}(\theta_i - \theta_j)$$

Across a line from i to j , the Power (P_{ij}) is equal to $(1/X_{ij})$ * difference in voltage phase angles

The DC Power Flow Approximation

$$P_{ij} = \frac{1}{X_{ij}}(\theta_i - \theta_j)$$

Across a line from i to j , the Power (P_{ij}) is equal to $(1/X_{ij})$ * difference in voltage phase angles

We can find X_{ij} using the conductor's reactance per km (standard value) and the distance between two buses.

This allows us to do many useful things. For example, if we treat the MVA line ratings of transmission lines as a Power (Watt) approximation, we can then find which lines are constrained if we know the Reactance and Phase difference

The DC Power Flow Approximation

DC Power Flow Approximation of AC Power $\leftrightarrow$ DC Power Flow (Ohm's Law)

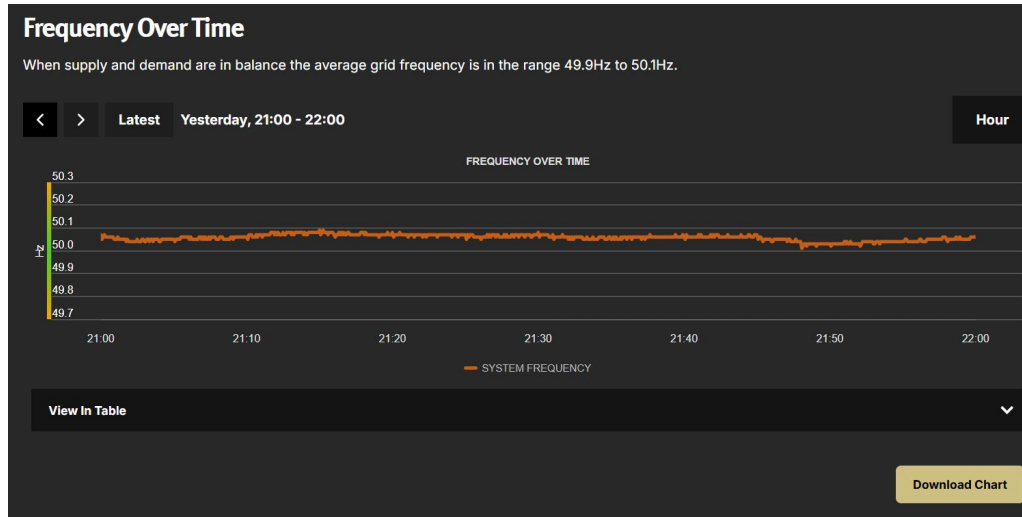
$$P_{ij} = \frac{1}{X_{ij}}(\theta_i - \theta_j) \quad \longleftrightarrow \quad I_{ij} = \frac{1}{R_{ij}}(V_i - V_j)$$

We are now done with DC Power Flow Approximation!

No need to understand the derivation, just use the P_{ij} formula like you would Ohm's law. Reactance (X) is essentially resistance, and voltage phase angle is essentially voltage. (Except it's not, but it actually works very well as an approximation, and is used by Eirgrid and other TSOs for a lot of power analysis).

Now for some more general but hopefully more interesting context

The Grid Operates at 50Hz



Look up “Eirgrid Smart Grid Dashboard” online. Eirgrid have an online tool which shows you lots of interesting data. One of them is the frequency of the grid. An awful lot of investment is put into keeping the AC frequency at 50Hz, ± 0.1 Hz.

Why?

Sudden changes of Frequency can damage equipment.

Small changes which are very rapid can be just as damaging as large changes over short times.

The metric RoCoF = Rate of Change of Frequency. You may come across it a lot and it's crucial for Eirgrid to minimise it.

Almost every piece of electrical equipment has predefined cutout points if it detects a steep RoCoF. This is what causes rolling blackouts.

This is how Spain and Portugal looked like from space during a massive blackout



Inertia and SNSP

Frequency on the grid is stabilised partly thanks to enormous spinning masses like Synchronous gas turbine generators.

Renewables do not provide inertia. They are considered “System-Non-Synchronous”
As we integrate more renewables into the grid, we reach higher levels of SNSP or “System-Non-Synchronous-Penetration”.

Reducing SNSP is a key reason behind
Curtailment of renewable generation



The National Control Center (NCC)

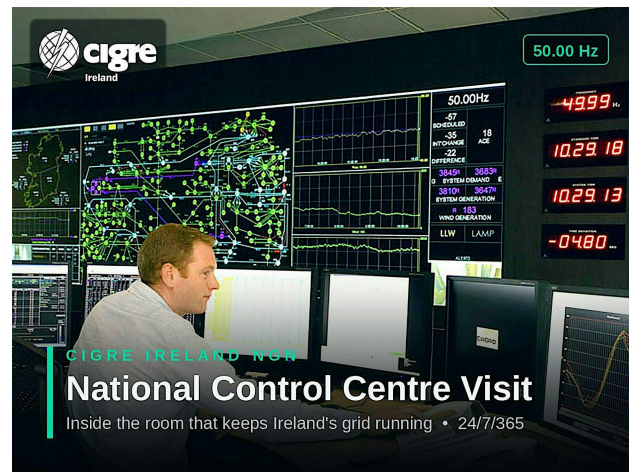


The NCC

Every country has an Eirgrid, and every Eirgrid has a National Control Centre (NCC)

Because the grid is a living system, a lot of very important decisions have to be made live in this room.

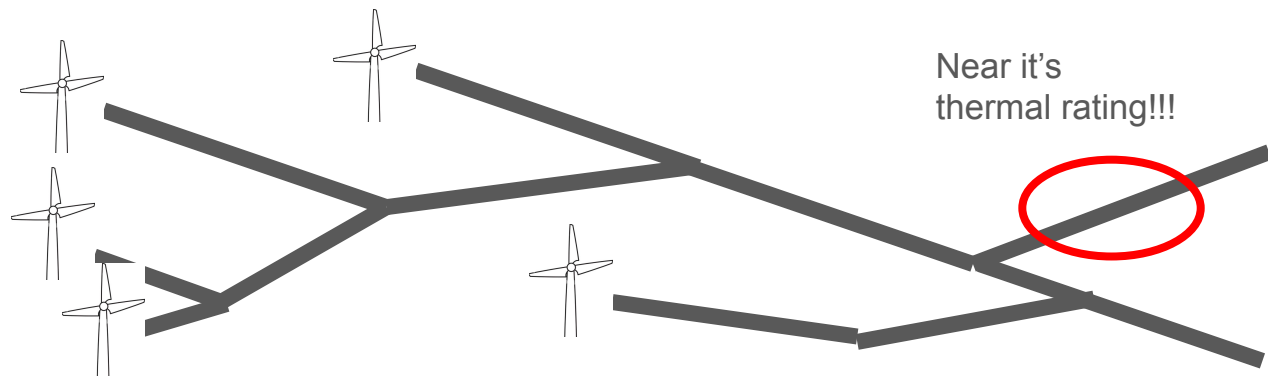
This is where a lot of constraint and curtailment decisions are ordered.



Constraint Groups

When a line is at risk of reaching its thermal rating, the NCC will constraint wind power to prevent this.

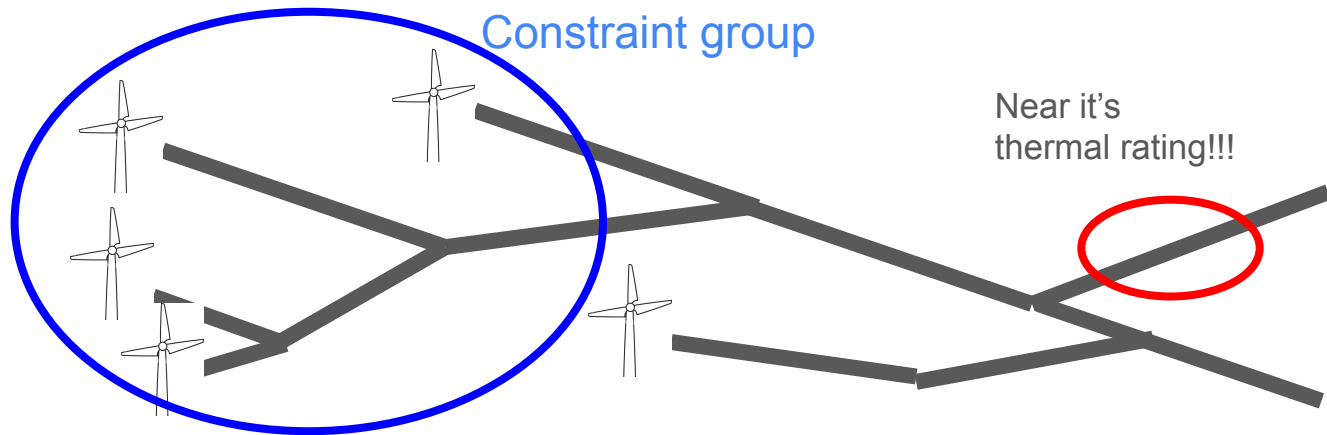
How do they know which farm to turn off?



Constraint Groups

When a line is at risk of reaching its thermal rating, the NCC will constraint wind power to prevent this.

How do they know which farm to turn off? - every farm in a certain group is turned off by the same proportion



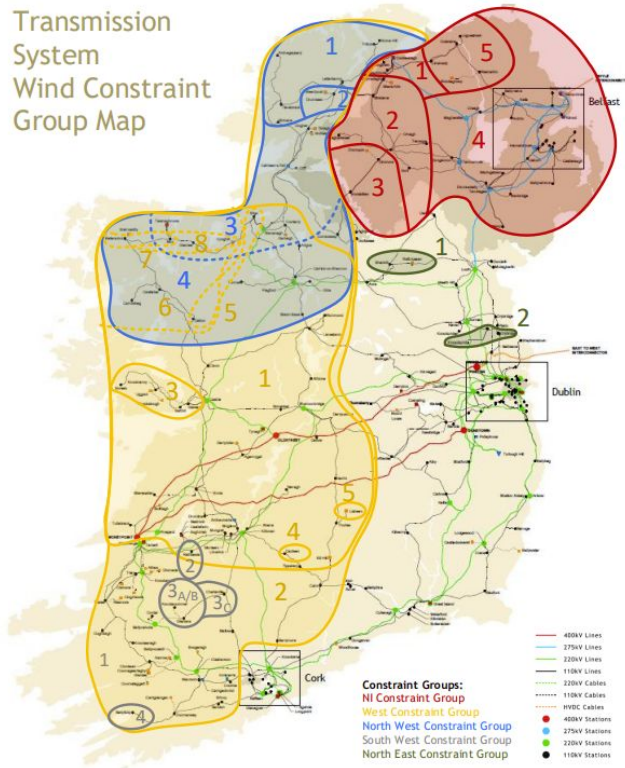
Constraint Groups

Constraint groups are used throughout Eirgrid for future forecasting, constraint ordering, and many more tasks.

But there is no one system to define a constraint group.

Sometimes constraint groups are defined on the fly.

And different departments have different tools which create and treat constraint groups differently



Creating a robust and simple method which takes little data and creates effective constraint groups could be an interesting problem.

A good constraint group has a high shift factor on the necessary transmission line, but also has farms of high physical closeness and has as large a constraint group as possible for fairness

Figure 2: North West Geographical Constraint Groups

Dynamic Line Rating (DLR)

Transmission lines are given conservative ratings based on thermal properties at different temperatures and currents.

However, the same current poses less of a risk at colder temperatures.

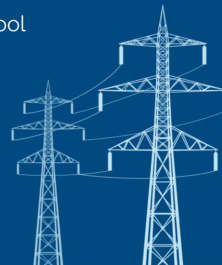
Clever solutions are being developed to monitor temperature and current and feedback line ratings in a useful way to prevent unnecessary constraint



BLOG POST:

Dynamic Line Ratings

An Innovative Tool
for Increasing
Grid Capacity

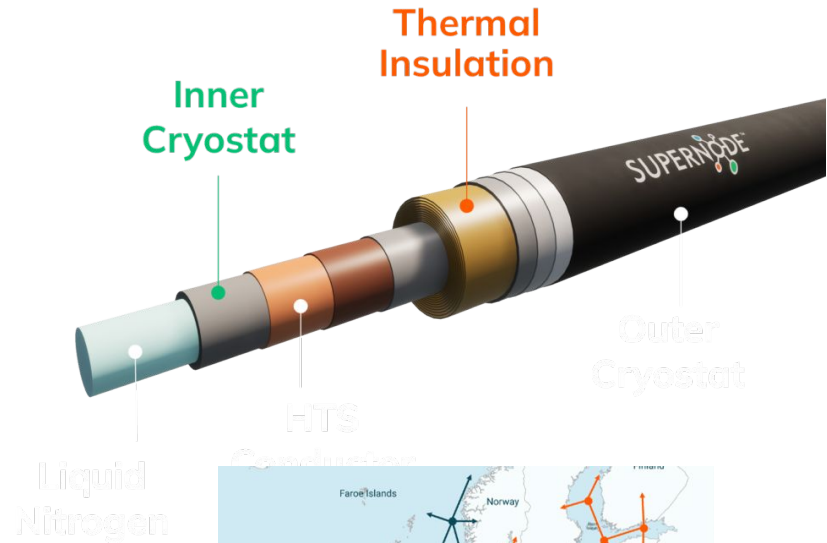


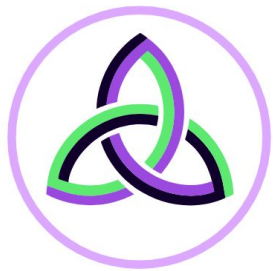
Superconducting Transmission

When certain conductors reach an extremely low temperature, their resistance becomes 0 and they become superconducting.

Superconducting Transmission has been used in Seoul in South Korea in an urban area to transmit high power densities underground

A [UCD Report](#) suggested that a pan-european “Supergrid” would could energy costs by 32%





Thank you!!

